Thesis — Detailed Table of Contents (Annotated)

Working title: Routine Pre-Announcement Disclosure and M&A Outcomes: Evidence from Form 10-K Narratives

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- 1.4.5 Interpretive discipline: associational evidence, mechanism tests, and causal claims
- 1.5 Routine Disclosure and Deal Premia
  - 1.5.1 Premium measurement and the interpretation of direction
  - 1.5.2 Information asymmetry and valuation uncertainty
  - 1.5.3 Processing-cost proxies: readability and disclosure volume
  - 1.5.4 Precision-oriented proxies: forward-looking disclosure and specificity
  - 1.5.5 Risk disclosure: risk transparency versus risk level
  - 1.5.6 Tone: news versus strategy
- 1.6 Routine Disclosure and Deal Completion
  - 1.6.1 Completion as a multi-channel outcome
  - 1.6.2 Information quality, renegotiation, and deal failure
  - 1.6.3 Regulatory scrutiny and completion frictions
  - 1.6.4 Deal communication and time-to-close
- 1.7 Moderators and Cross-Sectional Tests
  - 1.7.1 Payment method, valuation risk, and disclosure credibility
  - 1.7.2 Industry relatedness and the target information environment
  - 1.7.3 Regulatory exposure and institutional constraints
- 1.8 Research Gaps and Thesis Positioning
  - 1.8.1 Unresolved issues in existing evidence
  - 1.8.2 Measures, confounds, and planned empirical use
  - 1.8.3 Contribution statements and roadmap to hypotheses and design

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Chapter 2: Hypotheses and Research Design

Purpose: bridge the lit review to the empirics. States testable hypotheses, defines the identification strategy, specifies all econometric models.

- **2.1 Hypotheses Development**
  - 2.1.1 Disclosure quality and deal premia
    - H1a (information-risk pricing): better disclosure → higher premia (reduced protective discount)
    - H1b (precision/overpayment avoidance): better disclosure → lower premia (less overpayment)

        - Discuss which indices load on which hypothesis
    - 2.1.2 Disclosure quality and deal completion
        - H2: better disclosure → higher completion probability (reduced post-announcement frictions)
    - 2.1.3 Processing costs and deal outcomes
        - H3: higher processing costs → (a) lower premia, (b) lower completion probability
    - 2.1.4 Moderating effects
        - Payment method: cash vs. stock (differential valuation risk, strategic disclosure incentives)
        - Industry relatedness: cross-industry = more reliance on public disclosure
- **2.2 Empirical Strategy**
    - 2.2.1 Timing and identification
        - Most recent 10-K filed ≥90 days before announcement
        - Not causal; reduces mechanical endogeneity vs. deal-window text
        - Robustness lags: 60, 120 days (configurable via `pipeline_config.R`)
    - 2.2.2 Econometric specifications
        - Premium: OLS with FE (Equation 1), estimated via `fixest::feols()`
        - Completion: logit (Equation 2), estimated via `fixest::feglm()`
        - Duration: AFT log-normal (Equation 3), withdrawn = right-censored; Weibull and Cox PH for robustness
    - 2.2.3 Fixed effects and inference
        - Industry × year FE (2-digit SIC × announcement year)
        - Clustering at 2-digit SIC level
- **2.3 Variable Definitions**
    - 2.3.1 Dependent variables
        - Premium: SDC 1-day (primary), winsorised 1%/99%, bounded [−100%, 300%]; 1-week and 4-week for robustness
        - Completion: binary (completed = 1, withdrawn = 0)
        - Time-to-close: days from announcement to effective/withdrawal date, winsorised
    - 2.3.2 Disclosure quality indices (summary table mapping variable → construct → section → econometric role)
        - Core: operational specificity, FL sentence share, risk transparency, tone LM
        - Supplementary: FL precision share, commitment density, hedging density

- Processing costs: Gunning Fog, log MD&A length
- 2.3.3 Control variables
  - ln(deal value), % cash, tender dummy
  - Discussion of why parsimonious (no Compustat controls available in current pipeline)

Tables: Table 2.1 (variable-construct mapping, extends Table 1.2 from lit review with econometric role column)

Chapter 3: Data and Sample Construction

Purpose: describe data sources, sample selection, 10-K retrieval, and sample characteristics. All numbers come from the pipeline outputs.

- **3.1 M&A Deal Sample**
  - 3.1.1 Data source and initial universe
    - SDC Platinum / Refinitiv
    - US public targets, all deal types, no minimum deal value at extraction
    - Starting year 2006: Item 1A mandatory from fiscal years ending Dec 2005
  - 3.1.2 Sample restrictions
    - Sequential cascade with exact counts from `sample_cascade_blueprint.csv`
    - **Table 3.1**: restriction cascade (step, rule, N before, N removed, N after, retention %)
    - Base sample: 4,186 deals; premium subsample: 3,102; duration subsample: 3,181
  - 3.1.3 Premium measurement
    - SDC pre-calculated vs. manual: agree within ±0.1% for 99.9% of obs
    - Rationale for SDC premium (consistency, comparability, edge-case avoidance)
- **3.2 10-K Filing Identification and Retrieval**
  - 3.2.1 Target-to-CIK matching
    - External company-CIK database (independent public repository)
    - Tier 1: exact name match (SDC target_name vs. database name)
    - Tier 2: normalised name match (lowercase, strip punctuation, collapse whitespace)
    - No fuzzy/string-distance matching — deterministic at both tiers
    - Export to Excel → manual curation → re-import
    - Ticker recycling caveat and its partial mitigation via name-based matching
  - 3.2.2 Filing selection
    - SEC EDGAR submissions API; most recent 10-K ≥90 days pre-announcement

        - Max lookback 450 days; prefer 10-K over 10-K/A
        - **Figure 3.1**: histogram of filing lag distribution
    - 3.2.3 Section extraction
        - MD&A and Item 1A Risk Factors extracted separately
        - Validation: min 100 words, max 200,000/100,000 words per section
- **3.3 Descriptive Statistics**
    - 3.3.1 Deal characteristics
        - **Table 3.2**: Panel A — deal-level summary stats (premium, completion rate, time-to-close, deal value, payment, tender)
    - 3.3.2 Distribution by year and industry
        - **Table 3.3** or **Figure 3.2**: deals per year (bar chart or table), with completion rate overlay
        - **Table 3.4**: deals by macro industry
        - Comment: 80.5% completion rate, consistent with prior literature
    - 3.3.3 Filing and text characteristics
        - **Table 3.5**: filing lag, MD&A word count, Risk Factors word count, Fog, log length

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Chapter 4: Textual Analysis Methodology

Purpose: describe the NLP pipeline in enough detail that a reviewer can replicate it. Maps directly to Modules 1–6 of the R pipeline.

- **4.1 Text Preprocessing** *(maps to Module 1: `01_text_cleaning.R`)*
    - 4.1.1 EDGAR-aware cleaning
        - HTML/XBRL removal, header/footer stripping, boilerplate removal (max span 1,200 chars)
        - Financial pattern preservation: numbers, %, $, million/billion, quarters
    - 4.1.2 Section-level processing
        - MD&A and Risk Factors cleaned as separate text streams throughout
        - Word counts before/after cleaning; reduction ratios logged
- **4.2 Tokenization and Document-Feature Matrices** *(maps to Module 2: `02_tokenization_dtm.R`)*
    - 4.2.1 Tokenization strategy
        - `quanteda` pipeline: word-level tokenization, lowercasing

    - Dual-track stopword removal: Snowball list with finance-critical exceptions (modals, negation words)
    - No stemming/lemmatisation (per Loughran & McDonald 2016)
    - Operational bigram compounding (35 phrases from config)
  - 4.2.2 Document-feature matrices
    - Separate DFMs for MD&A and Risk Factors
    - TF-IDF weighting for Risk Factors DFM (upweights firm-specific risk language over boilerplate)
- **4.3 Disclosure Index Construction** *(maps to Module 3: `03_construct_indices.R`)*
  - 4.3.1 Operational specificity
    - Two components: numeric density (regex: %, $, million/billion per 1,000w) + operational compound density (35 bigrams per 1,000w)
    - Equal-weight combination (0.5/0.5) after component standardisation
    - Applied to MD&A
  - 4.3.2 Forward-looking intensity
    - Sentence-level classification: modal channel (34 terms) + regex channel (12 multi-word patterns)
    - FL sentence share = FL sentences / total sentences (MD&A)
    - FL precision share = FL sentences with numbers / FL sentences (supplementary)
  - 4.3.3 Risk transparency
    - TF-IDF-weighted Risk Factors DFM, partitioned by SIC2 × year peer groups
    - Peer-group centroid = mean TF-IDF vector
    - Risk transparency = 1 − cosine(firm vector, peer centroid)
    - Higher = more differentiated (firm-specific) risk disclosure
  - 4.3.4 Managerial tone
    - Loughran-McDonald positive/negative word lists on MD&A DFM
    - Tone_LM = (Pos − Neg) / (Pos + Neg + 1)
    - Supplementary: commitment density (strong modals/1000w), hedging density (weak modals/1000w)
  - 4.3.5 Processing cost measures
    - Gunning Fog Index (MD&A)
    - Log MD&A word count (after cleaning)

- **4.4 Normalisation** *(maps to Module 3, hierarchical fallback)*
  - Hierarchical z-score: SIC2 × year (if cell ≥ 10) → SIC2 → year → global
  - ~30% achieve preferred level; remainder at coarser fallback
  - Fallback level audited and reported per index per observation
  - For regressions: raw indices + FE (avoids double de-meaning vs. normalised)
  - Standardisation to z-scores for coefficient comparability
- **4.5 Validation** *(maps to Modules 4–5)*
  - 4.5.1 KWIC concordance validation
    - Keyword-in-context samples for each index (exported to `kwic_samples.xlsx`)
    - Negation-context KWIC for tone
    - Manual annotation fields for qualitative review
  - 4.5.2 Distributional properties and correlation structure
    - **Table 4.1**: pairwise correlation matrix of 4 core normalised indices
    - Histograms and Q-Q plots (appendix or inline figures)
    - No extreme discretisation or floor/ceiling effects
  - 4.5.3 Industry and time variation
    - ANOVA: all 4 raw indices significant across 2-digit SIC industries
    - Time trends: gradual increase in filing length and risk volume over sample period
    - **Figure 4.1**: time-trend plot of raw indices by year

Chapter 5: Empirical Results

Purpose: present all regression results with interpretation tied to the hypotheses. Tables populated from `premium_results.csv`, `completion_results.csv`, `duration_results.csv`.

- **5.1 Summary Statistics**
  - **Table 5.1**: full summary stats (Panels A–E: outcomes, core indices, supplementary, processing costs, controls)
    - Data from `econ_summary_stats.csv` — values already available
  - Commentary: negative mean tone (−0.40), FL share ~8%, risk transparency IQR 0.43–0.72
- **5.2 Disclosure Quality and Deal Premia**
  - **Table 5.2**: OLS regressions P1–P4
    - P1: core 4 + controls + year FE (N = 3,102)

- P2: core 4 + controls + ind × year FE (N = 2,745)
- P3: core 4 + supplementary + controls + year FE (N = 3,095)
- P4: core 4 + processing costs + controls + year FE (N = 3,102)
- Interpretation:
  - Operational specificity: robustly negative (−2.5 to −5.0 pp/SD), significant at 1% → supports H1b
  - FL sentence share: weakly positive in P1/P4 (marginal significance) → suggestive of H1a
  - Risk transparency: insignificant → countervailing forces cancel
  - Tone: insignificant → routine tone not informative conditional on precision + FE
  - Commitment density (P3): positive, significant → strong modal language → higher premia
  - Log MD&A length (P4): negative, significant at 1% → supports H3 (processing costs)

- **5.3 Disclosure Quality and Deal Completion**
  - **Table 5.3**: logit regressions C1–C4
    - C1: core 4 + controls + year FE (N = 4,186)
    - C2: core 4 + controls + ind × year FE
    - C3: extended + controls + year FE
    - C4: core 4 + processing costs + controls + year FE
  - Interpretation: to be written once coefficients are populated
- **5.4 Disclosure Quality and Time-to-Close**
  - **Table 5.4**: AFT models AFT1–AFT4
    - AFT1: core 4 + controls (log-normal)
    - AFT2: extended + controls (log-normal)
    - AFT3: core 4 + processing costs (log-normal)
    - AFT4: core 4 + controls (Weibull — distributional robustness)
  - Time ratios: exp(β) > 1 = longer duration
  - Cox PH with stratification by year as further robustness (withdrawn = censored)
  - Interpretation: to be written once coefficients are populated
- **5.5 Robustness Checks**
  - 5.5.1 Alternative premium windows
    - Replicate P1 with 1-week and 4-week SDC premia
    - Operational specificity result expected to survive
  - 5.5.2 Alternative filing lag thresholds

        - 60 and 120 days (configurable via `TIMING$lag_robustness` in config)
    - 5.5.3 Individual index regressions
        - **Table 5.5**: P5 one-at-a-time (from `premium_results.csv`, P5_* rows)
        - Operational specificity: −5.0 pp/SD standalone, significant at 1%
        - FL sentence share: +2.3 pp/SD standalone, significant at 1%
    - 5.5.4 Distributional robustness in duration models
        - Weibull vs. log-normal comparison
        - Cox PH with proper censoring of withdrawn deals

Chapter 6: Discussion and Conclusions

Purpose: interpret, contextualise, acknowledge limitations, suggest extensions. Short chapter (~5–7 pages).

- **6.1 Summary of Findings**
    - Restate key results: operational specificity negative on premia (robust), log length negative (processing cost), FL share weakly positive, tone and risk transparency insignificant
    - Completion and duration findings (once available)
- **6.2 Interpretation in Light of Competing Hypotheses**
    - Operational specificity → H1b dominant (precision/overpayment avoidance)
    - Consistent with Chen et al. (2025) on forward-looking MD&A
    - FL share → suggestive of H1a (information-risk pricing) for a different disclosure dimension
    - Tone null → credibility concerns, discounted by acquirers with private info
    - Risk transparency null → countervailing forces (transparency vs. risk level)
- **6.3 Contributions**
    - Section-specific routine-disclosure evidence (MD&A vs. Risk Factors, pre-announcement timing)
    - Construct-valid measurement with explicit mapping (Table 1.2 / Table 2.1)
    - Competing directional hypotheses with mechanism-oriented moderators
- **6.4 Limitations**
    - No causal identification (associational evidence)
    - Sparse SIC × year cells → only ~30% at preferred normalisation level
    - Bag-of-words / dictionary methods only (no contextual NLP)
    - US public targets with successful CIK match (potential selection)
    - Parsimonious controls (no Compustat financials in current pipeline)

- **6.5 Implications and Future Research**
  - Policy: disclosure quality has real consequences in market for corporate control
  - Extensions: Compustat controls, embedding/topic-model supplements, cross-country, joint analysis of routine + deal-specific disclosure

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Bibliography

(Combined references from all chapters, managed via BibLaTeX/BibTeX)

Appendix

- **A: Variable Definitions and Data Dictionary**
  - Complete variable definitions table (all panels: outcomes, indices, controls, FE)
  - Source and construction for each variable
- **B: Sample Restriction Cascade**
  - Full cascade from initial SDC extract through CIK matching and text extraction
  - Steps 1–7 (deal sample) + steps 8–10 (text sample)
- **C: Normalisation Fallback Statistics**
  - Fallback level distribution per index (from `fallback_usage_by_index.csv`)
- **D: Additional Figures**
  - Histograms and Q-Q plots for all indices
  - Correlation heatmap
- **E: Additional Regression Tables**
  - Individual index regressions (P5 full results)
  - Alternative premium windows
  - Cox PH robustness